

references

Chapter 1

- [1] Launchbury, John. 2020. “A DARPA Perspective on Artificial Intelligence.” https://www.youtube.com/watch?v=O01G3tSYpU&ab_channel=DARPAtv.
- [2] Yosinski, Jason, Jeff Clune, Yoshua Bengio, and Hod Lipson. 2014. “How Transferable Are Features in Deep Neural Networks?” In *Proceedings of the 27th International Conference on Neural Information Processing Systems (NIPS’14)*, 3320–3328. MIT Press. <https://arxiv.org/abs/1411.1792>.
- [3] Vaswani, Ashish, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, et al. 2017. “Attention Is All You Need.” In *Proceedings of the 31st International Conference on Neural Information Processing Systems (NIPS’17)*, 6000–6010. Curran Associates Inc. <https://arxiv.org/abs/1706.03762>.
- [4] Devlin, Jacob, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.” <https://arxiv.org/abs/1810.04805>.
- [5] Radford, Alec, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. “Language Models Are Unsupervised Multitask Learners.” <https://api.semanticscholar.org/CorpusID:160025533>.
- [6] OpenAI. 2023. *GPT-4 Technical Report*. <https://arxiv.org/abs/2303.08774>.
- [7] Rae, Jack W., Sebastian Borgeaud, Trevor Cai, Katie Millican, Jordan Hoffmann, Francis Song, John Aslanides, et al. 2021. “Scaling Language Models: Methods, Analysis & Insights from Training Gopher.” *arXiv preprint arXiv:2112.11446*. <https://arxiv.org/abs/2112.11446>.
- [8] Chowdhery, Aakanksha, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, et al. 2023. “PaLM: Scaling Language Modeling with Pathways.” *Journal of Machine Learning Research* 24 (1): 11324–11436. <https://dl.acm.org/doi/10.5555/3648699.3648939>.
- [9] Kaplan, Jared, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, et al. 2020. “Scaling Laws for Neural Language Models.” *arXiv preprint arXiv:2001.08361 [cs.LG]*. <https://arxiv.org/abs/2001.08361>.
- [10] Pan, Jeff Z., Diego Collarana, Maria Maleshkova, Patrick Westphal, and Hameedullah Khan. 2023. “Large Language Models and Knowledge Graphs: Opportunities and Challenges.” *Transactions on Graph Data and Knowledge* 1: 2:1–2:38. <https://doi.org/10.4230/TGDK.1.1.2>. <https://arxiv.org/abs/2308.06374>.
- [11] Pan, Shirui, Jie Zhou, Zonghan Wu, Chen Gong, Yu Sun, Yangqiu Song, Chengqi Zhang, et al. 2024. “Unifying Large Language Models and Knowledge Graphs: A Roadmap.” *IEEE*

- Transactions on Knowledge and Data Engineering* 36: 3580–3599. <https://ieeexplore.ieee.org/document/10387715>.
- [12] Grande, Daniel, Jorge Machado, Bryan Petzold, and Marcus Roth. 2020. “Reducing Data Costs without Jeopardizing Growth.” *McKinsey Digital*. <https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/reducing-data-costs-without-jeopardizing-growth>.
 - [13] Lecue, Freddy. 2020. “On the Role of Knowledge Graphs in Explainable AI.” *Semantic Web* 11 (1): 41–51. <http://semantic-web-journal.org/system/files/swj2259.pdf>.
 - [14] Zhang, Haoyu, Zhenghao Liu, Chenyan Xiong, and Zhiyuan Liu. 2020. “Grounded Conversation Generation as Guided Traverses in Commonsense Knowledge Graphs.” <https://arxiv.org/abs/1911.02707>.
 - [15] Barrasa, Jesús, Amy E. Hodler, and Jim Webber. 2021. *Knowledge Graphs*. O’Reilly Media. <https://www.oreilly.com/library/view/knowledge-graphs/9781098104863/>.

Chapter 2

- [1] Hulten, Geoff. 2018. *Building Intelligent Systems: A Guide to Machine Learning Engineering*. Apress.
- [2] Molina, Miguel. 2020. “What Is an Intelligent System?” <https://arxiv.org/abs/2009.09083>.
- [3] Negro, Alessandro. 2021. *Graph-Powered Machine Learning*. Manning Publications.
- [4] Wu, Zhilin, et al. 2023. “Reasoning or Reciting? Exploring the Capabilities and Limitations of Language Models Through Counterfactual Tasks.” <https://arxiv.org/abs/2307.02477>.
- [5] Dong, Xin, et al. 2014. “Knowledge Vault: A Web-Scale Approach to Probabilistic Knowledge Fusion.” In *Proceedings of SIGKDD*, 601–610. ACM.
- [6] Nickel, Maximilian, Kevin Murphy, Volker Tresp, and Evgeniy Gabrilovich. 2016. “A Review of Relational Machine Learning for Knowledge Graphs.” *Proceedings of the IEEE* 104 (1): 11–33.
- [7] Wang, Quan, Zhendong Mao, Bin Wang, and Li Guo. 2017. “Knowledge Graph Embedding: A Survey of Approaches and Applications.” *IEEE Transactions on Knowledge and Data Engineering* 29 (12): 2724–2743.
- [8] Hogan, Aidan, et al. 2020. “Knowledge Graphs.” <https://arxiv.org/abs/2003.02320>.
- [9] Stokman, Frans N., and Pieter H. de Vries. 1988. “Structuring Knowledge in a Graph.” In *Human-Computer Interaction*, 186–206. Springer.
- [10] Sculley, David, Gary Holt, Daniel Golovin, Eugene Davydov, Todd Phillips, Dietmar Ebner, Vinay Chaudhary, et al. 2015. “Hidden Technical Debt in Machine Learning Systems.” In *Proceedings of the 28th International Conference on Neural Information Processing Systems (NIPS’15)*, vol. 2, 2503–2511. MIT Press.
- [11] Pan, Jeff Z., Diego Collarana, Maria Maleshkova, Patrick Westphal, and Hameedullah Khan. 2023. “Large Language Models and Knowledge Graphs: Opportunities and Challenges.” *Transactions on Graph Data and Knowledge* 1 (1): 2:1–2:38. <https://arxiv.org/abs/2308.06374>.
- [12] Wirth, Rüdiger, and Jochen Hipp. 2000. “CRISP-DM: Towards a Standard Process Model for Data Mining.” In *Proceedings of the Fourth International Conference on the Practical Application of Knowledge Discovery and Data Mining*, 29–39. ACM.

Chapter 3

- [1] Köhler, Sebastian, Melissa Gargano, Nico Matentzoglou, et al. 2021. “The Human Phenotype Ontology in 2021.” *Nucleic Acids Research* 49 (D1): D1207–D1217. <https://academic.oup.com/nar/article/49/D1/D1207/6017351>.
- [2] Orlandi, Fabrizio, Damien Graux, and Declan O’Sullivan. 2021. “Benchmarking RDF Metadata Representations: Reification, Singleton Property, and RDF.” In *2021 IEEE 15th International*

Conference on Semantic Computing (ICSC), 233–240. IEEE. <https://ieeexplore.ieee.org/document/9364401/>.

Chapter 4

- [1] Stickle, David, and John Turk. 1997. “A Kinetic Mass Balance Model for 1, 5-Anhydroglucitol: Applications to Monitoring of Glycemic Control.” *American Journal of Physiology—Endocrinology and Metabolism* 273 (4): E821–E830.
- [2] Nicholson, David N., and Casey S. Greene. 2020. “Constructing Knowledge Graphs and Their Biomedical Applications.” *Computational and Structural Biotechnology Journal* 18: 1414–1428. <https://doi.org/10.1016/j.csbj.2020.05.017>.
- [3] Himmelstein, Daniel Scott, Antoine Lizée, Christine Hessler, Leo Brueggeman, Sabrina L. Chen, Dexter Hadley, et al. 2017. “Systematic Integration of Biomedical Knowledge Prioritizes Drugs for Repurposing.” *eLife* 6: e26726. <https://doi.org/10.7554/eLife.26726>.
- [4] Choi, Edward, Mohammad Taha Bahadori, Le Song, Walter F. Stewart, and Jimeng Sun. 2016. “GRAM: Graph-Based Attention Model for Healthcare Representation Learning.” <https://arxiv.org/abs/1611.07012v3>.
- [5] Shen, Zhen, You-Hua Zhang, Kyungsook Han, Asoke K. Nandi, Barry Honig, and De-Shuang Huang. 2017. “miRNA-Disease Association Prediction with Collaborative Matrix Factorization.” *Complexity* 2017: 2498957. <https://doi.org/10.1155/2017/2498957>.
- [6] Crichton, Gamal, Yufan Guo, Sampo Pyysalo, and Anna Korhonen. 2018. “Neural Networks for Link Prediction in Realistic Biomedical Graphs: A Multidimensional Evaluation of Graph Embedding-Based Approaches.” *BMC Bioinformatics* 19 (1): 176. <https://doi.org/10.1186/s12859-018-2163-9>.
- [7] Dimitrakopoulos, Christos, Sravanth Kumar Hindupur, Luca Häfliger, Jonas Behr, Hesam Montazeri, Michael N. Hall, and Niko Beerenwinkel. 2018. “Network-Based Integration of Multi-Omics Data for Prioritizing Cancer Genes.” *Bioinformatics* 34 (13): 2201–2209. <https://doi.org/10.1093/bioinformatics/bty148>.
- [8] Wang, Meng, Mengyue Liu, Jun Liu, Sen Wang, Guodong Long, and Buyue Qian. 2017. “Safe Medicine Recommendation via Medical Knowledge Graph Embedding.” <https://arxiv.org/abs/1710.05980v2>.
- [9] Shang, Junyuan, Cao Xiao, Tengfei Ma, Hongyan Li, and Jimeng Sun. 2019. “GAMENet: Graph Augmented Memory Networks for Recommending Medication Combination.” *Proceedings of the AAAI Conference on Artificial Intelligence* 33 (1): 1126–1133. <https://doi.org/10.1609/aaai.v33i01.33011126>.
- [10] Yang, Kuo, Ning Wang, Guangming Liu, Ruyu Wang, Jian Yu, Runshun Zhang, et al. 2018. “Heterogeneous Network Embedding for Identifying Symptom Candidate Genes.” *Journal of the American Medical Informatics Association* 25 (12): 1541–1549. <https://doi.org/10.1093/jamia/ocy117>.
- [11] Wang, Hua, Heng Huang, Chris Ding, and Feiping Nie. 2013. “Predicting Protein–Protein Interactions from Multimodal Biological Data Sources via Nonnegative Matrix Tri-Factorization.” *Journal of Computational Biology* 20 (4): 344–358. <https://doi.org/10.1089/cmb.2012.0273>.
- [12] Wu, Qingyao, Zhenyu Wang, Chunshan Li, Yunming Ye, Yueping Li, and Ning Sun. 2015. “Protein Functional Properties Prediction in Sparsely-Label PPI Networks through Regularized Nonnegative Matrix Factorization.” *BMC Systems Biology* 9 (Suppl 1): S9. <https://doi.org/10.1186/1752-0509-9-s1-s9>.
- [13] Agrawal, Monica, Marinka Zitnik, and Jure Leskovec. 2018. “Large-Scale Analysis of Disease Pathways in the Human Interactome.” *Pacific Symposium on Biocomputing*, 111–122. World Scientific.
- [14] Menche, Jörg, et al. 2015. “Uncovering Disease–Disease Relationships through the Incomplete Interactome.” *Science* 347 (6224): 1257601.

- [15] Chatr-Aryamontri, Andrew, et al. 2015. "The BioGRID Interaction Database: 2015 Update." *Nucleic Acids Research* 43 (D1): D470–D478.
- [16] Blondel, Vincent D., Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. 2008. "Fast Unfolding of Communities in Large Networks." *Journal of Statistical Mechanics: Theory and Experiment* 2008 (10): P10008.
- [17] Schaeffer, Satu Elisa. 2007. "Graph Clustering." *Computer Science Review* 1 (1): 27–64.
- [18] DiMasi, Joseph A., Henry G. Grabowski, and Ronald W. Hansen. 2016. "Innovation in the Pharmaceutical Industry: New Estimates of R&D Costs." *Journal of Health Economics* 47: 20–33. <https://doi.org/10.1016/j.jhealeco.2016.01.012>.
- [19] Reichert, Janice M. 2003. "Trends in Development and Approval Times for New Therapeutics in the United States." *Nature Reviews Drug Discovery* 2 (9): 695–702. <https://doi.org/10.1038/nrd1178>.
- [20] Hay, Michael, David W. Thomas, John L. Craighead, C. Economides, and Jesse Rosenthal. 2014. "Clinical Development Success Rates for Investigational Drugs." *Nature Biotechnology* 32 (1): 40–51. <https://doi.org/10.1038/nbt.2786>.
- [21] Abdelaziz, Ibrahim, Achille Fokoue, Oktie Hassanzadeh, Ping Zhang, and Mohammad Sadoghi. 2017. "Large-Scale Structural and Textual Similarity-Based Mining of Knowledge Graph to Predict Drug–Drug Interactions." *Journal of Web Semantics* 44: 104–117. <https://doi.org/10.1016/j.websem.2017.06.002>.
- [22] Dai, Wen, Xi Liu, Yibo Gao, Lin Chen, Jianglong Song, Di Chen, et al. 2015. "Matrix Factorization-Based Prediction of Novel Drug Indications by Integrating Genomic Space." *Computational and Mathematical Methods in Medicine* 2015: 275045. <https://doi.org/10.1155/2015/275045>.
- [23] Himmelstein, Daniel Scott, Antoine Lizee, Christine Hessler, Leo Brueggeman, Sabrina L. Chen, Dexter Hadley, et al. 2017. "Systematic Integration of Biomedical Knowledge Prioritizes Drugs for Repurposing." *eLife* 6: e26726. <https://doi.org/10.7554/eLife.26726>.
- [24] Ezzat, Ali, Peilin Zhao, Min Wu, Xiao-Li Li, and Chee-Keong Kwoh. 2017. "Drug–Target Interaction Prediction with Graph Regularized Matrix Factorization." *IEEE/ACM Transactions on Computational Biology and Bioinformatics* 14 (3): 646–656. <https://doi.org/10.1109/tcbb.2016.2530062>.
- [25] Eslami Manoochchhri, Hafez, and Mehrdad Nourani. 2018. "Predicting Drug–Target Interaction Using Deep Matrix Factorization." In *2018 IEEE Biomedical Circuits and Systems Conference (BioCAS)*, 1–4. <https://doi.org/10.1109/biocas.2018.8584817>.
- [26] Wang, Yin-Ying, Chunfeng Cui, Liqun Qi, Hong Yan, and Xing-Ming Zhao. 2019. "DrPOCS: Drug Repositioning Based on Projection onto Convex Sets." *IEEE/ACM Transactions on Computational Biology and Bioinformatics* 16 (4): 1360–1372. <https://doi.org/10.1109/tcbb.2018.2830384>.
- [27] Liu, Yong, Min Wu, Chunyan Miao, Peilin Zhao, and Xiao-Li Li. 2016. "Neighborhood Regularized Logistic Matrix Factorization for Drug–Target Interaction Prediction." *PLoS Computational Biology* 12 (2): e1004760. <https://doi.org/10.1371/journal.pcbi.1004760>.
- [28] Himmelstein, Daniel S., and Sergio E. Baranzini. 2015. "Heterogeneous Network Edge Prediction: A Data Integration Approach to Prioritize Disease-Associated Genes." *PLoS Computational Biology* 11 (4): e1004259. <https://doi.org/10.1371/journal.pcbi.1004259>.
- [29] Sun, Yizhou, Rick Barber, Manish Gupta, Charu C. Aggarwal, and Jiawei Han. 2009. "Co-Author Relationship Prediction in Heterogeneous Bibliographic Networks." In *2009 Ninth IEEE International Conference on Data Mining*, 1170–1175. IEEE. <https://doi.org/10.1109/ICDM.2009.36>.
- [30] Stephens, Matthew, and David J. Balding. 2009. "Bayesian Statistical Methods for Genetic Association Studies." *Nature Reviews Genetics* 10 (10): 681–690. <https://doi.org/10.1038/nrg2615>.
- [31] Sawcer, Stephen. 2008. "The Complex Genetics of Multiple Sclerosis: Pitfalls and Prospects." *Brain* 131 (12): 3118–3131. <https://doi.org/10.1093/brain/awn081>.
- [32] Santos, Alberto, Ana R. Colaço, Annelaura B. Nielsen, Lili Niu, Maximilian Strauss, Philipp E. Geyer, et al. 2022. "A Knowledge Graph to Interpret Clinical Proteomics Data." *Nature Biotechnology* 40:692–702.

- [33] Rotmensch, Maya, Yoni Halpern, Abdulhakim Tlimat, Steven Horng, and David Sontag. 2017. “Learning a Health Knowledge Graph from Electronic Medical Records.” *Scientific Reports* 7 (1): 5994.
- [34] Barton, Emma, Tony Freeman, Fran Baum, Shadi Javanparast, and Allan Lawless. 2019. “The Feasibility and Potential Use of Case-Tracked Client Journeys in Primary Healthcare: A Pilot Study.” *BMJ Open* 9 (5): e024419. <https://doi.org/10.1136/bmjopen-2018-024419>.
- [35] Hipp, Rachel, Erik Abel, and Robert J. Weber. 2016. “A Primer on Clinical Pathways.” *Hospital Pharmacy* 51 (5): 416–421. <https://doi.org/10.1310/hpj5105-416>.
- [36] Chang, J. Morris, Di Zhuang, and G. Dumindu Samaraweera. 2023. *Privacy-Preserving Machine Learning*. Manning.

Chapter 5

- [1] Chernow, Ron. 1998. *Titan: The Life of John D. Rockefeller, Sr.* Random House.
- [2] Vaswani, Ashish, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, et al. 2017. “Attention Is All You Need.” <https://arxiv.org/abs/1706.03762>.
- [3] Kaplan, Jared, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, et al. 2020. “Scaling Laws for Neural Language Models.” <https://arxiv.org/abs/2001.08361>.

Chapter 7

- [1] European Centre for Disease Prevention and Control. 2016. *Communicable Disease Threats Report*.
- [2] Shi, Jialu, Zhiwei Yuan, Wenjie Guo, Chao Ma, Jun Chen, and Ming Zhang. 2023. “Knowledge-Graph-Enabled Biomedical Entity Linking: A Survey.” *World Wide Web* 26: 1–30. <https://link.springer.com/article/10.1007/s11280-023-01144-4>.
- [3] Bodenreider, Olivier. 2004. “The Unified Medical Language System (UMLS): Integrating Biomedical Terminology.” *Nucleic Acids Research* 32 (suppl_1): D267–D270. https://academic.oup.com/nar/article-abstract/32/suppl_1/D267/2505235.
- [4] Neumann, Mark, Daniel King, Iz Beltagy, and Waleed Ammar. 2019. “scispaCy: Fast and Robust Models for Biomedical Natural Language Processing.” <https://arxiv.org/abs/1902.07669>.
- [5] Cornet, Ronald, and Nicolette de Keizer. 2008. “Forty Years of SNOMED: A Literature Review.” *BMC Medical Informatics and Decision Making* 8 (Suppl 1): S2. <https://bmcmmedinformdecismak.biomedcentral.com/articles/10.1186/1472-6947-8-S1-S2>.
- [6] European Commission. 2022. “Proposal for a Regulation: Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act).” <https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:52022PC0338>.
- [7] Widenfalk, Bengt, Jonas Rastad, Stina Wilbrand, and Bengt Vinnars. 2014. *Guide to the Quality and Safety of Tissues and Cells for Human Application*. 2nd Ed. European Directorate for the Quality of Medicines & HealthCare.
- [8] Köhler, Sebastian, Melissa Gargano, Nico Matentzoglou, Laura C. Carmody, David Lewis-Smith, Nicole A. Vasilevsky, et al. 2021. “The Human Phenotype Ontology in 2021.” *Nucleic Acids Research* 49 (D1): D1207–D1217.

Chapter 9

- [1] Negro, Alessandro. 2021. *Graph-Powered Machine Learning*. Manning Publications.
- [2] Mukkamala, Anoop, Peter Szolovits, and Judy Lin. 2021. “Representation Learning: From Diagnoses to Healthcare Analytics.” *Data Science* 4 (2): 125–146. <https://content.iospress.com/articles/data-science/ds007>.
- [3] Ma, Yao, and Jie Tang. 2021. *Deep Learning on Graphs*. Cambridge University Press.

- [4] Hamilton, William L. 2020. *Graph Representation Learning*. Morgan & Claypool.
- [5] Kipf, Thomas N., and Max Welling. 2016. “Semi-Supervised Classification with Graph Convolutional Networks.” In *International Conference on Learning Representations (ICLR 2016)*. <https://arxiv.org/abs/1609.02907>.
- [6] McPherson, Miller, Lynn Smith-Lovin, and James M. Cook. 2001. “Birds of a Feather: Homophily in Social Networks.” *Annual Review of Sociology* 27 (1): 415–444.
- [7] Zhou, Dengyong, Olivier Bousquet, Thomas Lal, Jason Weston, and Bernhard Schölkopf. 2004. “Learning with Local and Global Consistency.” In *Proceedings of the 17th International Conference on Neural Information Processing Systems (NeurIPS 2004)*. <https://proceedings.neurips.cc/paper/2003/hash/87682805257e619d49b8e0dfdc14affa-Abstract.html>.
- [8] Donnat, Claire, Marinka Zitnik, David Hallac, and Jure Leskovec. 2018. “Graph Wavelets for Structural Role Similarity in Complex Networks.” In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD 2018)*, 373–382. <https://dl.acm.org/doi/10.1145/3219819.3220025>.
- [9] Yang, Zhilin, William Cohen, and Ruslan Salakhutdinov. 2016. “Revisiting Semi-Supervised Learning with Graph Embeddings.” In *Proceedings of the 33rd International Conference on Machine Learning (ICML 2016)*, 40–48. <https://proceedings.mlr.press/v48/yanga16.html>.
- [10] Tang, Lei, and Huan Liu. 2009. “Relational Learning via Latent Social Dimensions.” In *Proceedings of the 15th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD 2009)*, 817–826. <https://dl.acm.org/doi/10.1145/1557019.1557109>.
- [11] Adamic, Lada A., and Eytan Adar. 2003. “Friends and Neighbors on the Web.” *Social Networks* 25 (3): 211–230.
- [12] Ying, Rex, Ruining He, Kaifeng Chen, Pong Eksombatchai, William L. Hamilton, and Jure Leskovec. 2018. “Graph Convolutional Neural Networks for Web-Scale Recommender Systems.” In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD 2018)*, 974–983. <https://dl.acm.org/doi/10.1145/3219819.3219890>.
- [13] Nickel, Maximilian, Kevin Murphy, Volker Tresp, and Evgeniy Gabrilovich. 2015. “A Review of Relational Machine Learning for Knowledge Graphs.” *Proceedings of the IEEE* 104 (1): 11–33. <https://ieeexplore.ieee.org/document/7358050>.
- [14] Zitnik, Marinka, Monica Agrawal, and Jure Leskovec. 2018. “Modeling Polypharmacy Side Effects with Graph Convolutional Networks.” *Bioinformatics* 34 (13): i457–i466.
- [15] Bordes, Antoine, Nicolas Usunier, Alberto Garcia-Duran, Jason Weston, and Oksana Yakhnenko. 2013. “Translating Embeddings for Modeling Multi-Relational Data.” In *Proceedings of the 26th International Conference on Neural Information Processing Systems (NeurIPS 2013)*, 2787–2795. https://papers.nips.cc/paper_files/paper/2013/hash/1cecc7a77928ca8133fa24680a88d2f9-Abstract.html.
- [16] Wang, Hua, Heng Huang, Chris Ding, and Feiping Nie. 2013. “Predicting Protein–Protein Interactions from Multimodal Biological Data Sources via Nonnegative Matrix Tri-Factorization.” *Journal of Computational Biology* 20 (4): 344–358. <https://doi.org/10.1089/cmb.2012.0273>.
- [17] Berlusconi, Giulia, Francesco Calderoni, Nicola Parolini, Marco Verani, and Corrado Piccardi. 2016. “Link Prediction in Criminal Networks: A Tool for Criminal Intelligence Analysis.” *PLoS ONE* 11 (4): e0154244. <https://doi.org/10.1371/journal.pone.0154244>.
- [18] Agrawal, Monica, Marinka Zitnik, and Jure Leskovec. 2018. “Large-Scale Analysis of Disease Pathways in the Human Interactome.” In *Pacific Symposium on Biocomputing (PSB)*, 111–122. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5731453/>.
- [19] Pandit, Shashank, Duen Horng Chau, Saikat Wang, and Christos Faloutsos. 2007. “NetProbe: A Fast and Scalable System for Fraud Detection in Online Auction Networks.” In *Proceedings of the 16th International World Wide Web Conference (WWW)*, 201–210. <https://www.cs.cmu.edu/~christos/PUBLICATIONS/netprobe-www07.pdf>.

- [20] Raghavan, Usha Nandini, Reka Albert, and Soundar Kumara. 2007. “Near Linear Time Algorithm to Detect Community Structures in Large-Scale Networks.” *Physical Review E* 76 (3): 036106. <https://arxiv.org/pdf/0709.2938>.
- [21] Gilmer, Justin, Samuel S. Schoenholz, Patrick F. Riley, Oriol Vinyals, and George E. Dahl. 2017. “Neural Message Passing for Quantum Chemistry.” In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 1263–1272. <https://proceedings.mlr.press/v70/gilmer17a.html>.
- [22] Li, Yujia, Chenjie Gu, Thomas Dullien, Oriol Vinyals, and Pushmeet Kohli. 2019. “Graph Matching Networks for Learning the Similarity of Graph Structured Objects.” In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 3835–3845. <https://proceedings.mlr.press/v97/li19d.html>.
- [23] Sen, Prithviraj, Galileo Namata, Mustafa Bilgic, Lise Getoor, Brian Galligher, and Tina Eliassi-Rad. 2008. “Collective Classification in Network Data.” *AI Magazine* 29 (3): 93–106. <https://doi.org/10.1609/aimag.v29i3.2157>.
- [24] Zachary, Wayne W. 1977. “An Information Flow Model for Conflict and Fission in Small Groups.” *Journal of Anthropological Research* 33 (4): 452–473. <https://www.jstor.org/stable/3629752>.
- [25] Grover, Aditya, and Jure Leskovec. 2016. “node2vec: Scalable Feature Learning for Networks.” In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 855–864. <https://arxiv.org/abs/1607.00653>.

Chapter 10

- [1] Negro, Alessandro. 2021. *Graph-Powered Machine Learning*. Manning Publications.
- [2] Fouss, François, Marco Saerens, and Masashi Shimbo. 2016. *Algorithms and Models for Network Data and Link Analysis*. Cambridge University Press. <https://doi.org/10.1017/CBO9781316418321>.
- [3] Cormen, Thomas H., Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein. 2022. *Introduction to Algorithms*. 4th ed. MIT Press.
- [4] Baesens, Bart, Véronique Van Vlasselaer, and Wouter Verbeke. 2015. *Fraud Analytics Using Descriptive, Predictive, and Social Network Techniques: A Guide to Data Science for Fraud Detection*. Wiley.
- [5] Himmelstein, Daniel S., Antoine Lizée, Christine Hessler, et al. 2017. “Systematic Integration of Bio-medical Knowledge Prioritizes Drugs for Repurposing.” *eLife* 6: e26726. <https://doi.org/10.7554/eLife.26726>.
- [6] Himmelstein, Daniel S. 2016. “Our Hetnet Edge Prediction Methodology: The Modeling Framework for Project Rephetio.” *ThinkLab*. <https://doi.org/10.15363/thinklab.d210>.
- [7] Himmelstein, Daniel S. 2017. “D. Himmel/Learn V1.0: The Machine Learning Repository for Project Rephetio.” *Zenodo*. <https://doi.org/10.5281/zenodo.268654>.
- [8] Henderson, Keith, Brian Gallagher, Lei Li, Leman Akoglu, Tina Eliassi-Rad, Hanghang Tong, and Christos Faloutsos. 2011. “It’s Who You Know: Graph Mining Using Recursive Structural Features.” In *Proceedings of the 17th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 663–671. <https://doi.org/10.1145/2020408.2020512>.

Chapter 11

- [1] Ma, Yao, and Jie Tang. 2021. *Deep Learning on Graphs*. Cambridge University Press. <https://doi.org/10.1017/9781108924184>.
- [2] Hamilton, William L. 2020. *Graph Representation Learning*. Morgan & Claypool. <https://doi.org/10.2200/S01045ED1V01Y202009AIM046>.
- [3] Chami, Ines, Sami Abu-El-Hajja, Bryan Perozzi, Christopher Ré, and Kevin Murphy. 2022. “Machine Learning on Graphs: A Model and Comprehensive Taxonomy.” <https://arxiv.org/abs/2204.06025>.

- [4] Grover, Aditya, and Jure Leskovec. 2016. “node2vec: Scalable Feature Learning for Networks.” In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 855–864. <https://doi.org/10.1145/2939672.2939754>.
- [5] Bordes, Antoine, Nicolas Usunier, Alberto Garcia-Duran, Jason Weston, and Oksana Yakhnenko. 2013. “Translating Embeddings for Modeling Multi-Relational Data.” In *Advances in Neural Information Processing Systems 26 (NeurIPS 2013)*, 2787–2795. <https://proceedings.neurips.cc/paper/2013/hash/1cecc7a77928ca8133fa24680a88d2f9-Abstract.html>.
- [6] Nickel, Maximilian, Volker Tresp, and Hans-Peter Kriegel. 2011. “A Three-Way Model for Collective Learning on Multi-Relational Data.” In *Proceedings of the 28th International Conference on Machine Learning (ICML 2011)*, 809–816. https://icml.cc/2011/papers/438_icmlpaper.pdf.
- [7] Yang, Bishan, Wen-tau Yih, Xiaodong He, Jianfeng Gao, and Li Deng. 2015. “Embedding Entities and Relations for Learning and Inference in Knowledge Bases.” In *International Conference on Learning Representations (ICLR 2015)*. <https://arxiv.org/abs/1412.6575>.
- [8] Trouillon, Théo, Johannes Welbl, Sebastian Riedel, Éric Gaussier, and Guillaume Bouchard. 2016. “Complex Embeddings for Simple Link Prediction.” In *Proceedings of the 33rd International Conference on Machine Learning (ICML 2016)*, 2071–2080. <https://proceedings.mlr.press/v48/trouillon16.html>.
- [9] Kipf, Thomas N., and Max Welling. 2017. “Semi-Supervised Classification with Graph Convolutional Networks.” In *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/1609.02907>.
- [10] Veličković, Petar, Guillem Cucurull, Arantxa Csanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. 2018. “Graph Attention Networks.” In *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/1710.10903>.
- [11] Hamilton, William L., Rex Ying, and Jure Leskovec. 2017. “Inductive Representation Learning on Large Graphs.” In *Advances in Neural Information Processing Systems 30*, 1024–1034. <https://proceedings.neurips.cc/paper/2017/hash/5dd9db5e033da9c6fb5ba83c7a7e9bea9-Abstract.html>.
- [12] Jin, Bowen, Chi Han, Gang Liu, Meng Jiang, Heng Ji, and Jiawei Han. 2024. “Large Language Models on Graphs: A Comprehensive Survey.” <https://arxiv.org/abs/2312.02783>.

Chapter 12

- [1] Hamilton, William L., Zhitao Ying, and Jure Leskovec. 2017. “Inductive Representation Learning on Large Graphs.” *Advances in Neural Information Processing Systems* 30.
- [2] Kipf, Thomas N., and Max Welling. 2016. “Semi-Supervised Classification with Graph Convolutional Networks.” <https://arxiv.org/abs/1609.02907>.
- [3] Veličković, Petar, Guillem Cucurull, Arantxa Csanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. 2017. “Graph Attention Networks.” <https://arxiv.org/abs/1710.10903>.

Chapter 13

- [1] Lanham, Michael. 2025. *AI Agents in Action*. Manning Publications.
- [2] Gao, Yunfan, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, et al. 2023. “Retrieval Augmented Generation for Large Language Models: A Survey.” <https://arxiv.org/abs/2312.10997>.
- [3] Liu, Nelson F., Kevin Lin, John Hewitt, Aditi Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2023. “Lost in the Middle: How Language Models Use Long Contexts.” <https://arxiv.org/abs/2307.03172>.
- [4] Pan, Shirui, Linhao Luo, Yufei Wang, Chen Chen, Jiapu Wang, and Xindong Wu. 2023. “Unifying Large Language Models and Knowledge Graphs: A Roadmap.” <https://arxiv.org/abs/2306.08302>.

- [5] Jin, Bowen, Gang Liu, Chi Han, Meng Jiang, Heng Ji, and Jiawei Han. 2023. “Large Language Models on Graphs: A Comprehensive Survey.” <https://arxiv.org/abs/2312.02783>.
- [6] Yao, Shunyu, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2022. “ReAct: Synergizing Reasoning and Acting in Language Models.” <https://arxiv.org/abs/2210.03629>.
- [7] Bratanič, Tomaž, and Oskar Hane. 2025. *Knowledge Graph-Enhanced RAG*. Manning Publications.

Chapter 14

- [1] Neo4j. “Neo4j APOC Library: apoc.meta.schema.” <https://neo4j.com/docs/apoc/current/overview/apoc.meta/apoc.meta.schema/>.
- [2] Wei, Jason, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Ed Chi, Quoc Le, and Denny Zhou. 2023. “Chain of Thought Prompting Elicits Reasoning in Large Language Models.” <https://arxiv.org/abs/2201.11903>.
- [3] Nye, Maxwell, Anders Andreassen, Guy Gur-Ari, Henryk Michalewski, Joshua Austin, Daniel Bieber, et al. 2021. “Show Your Work: Scratchpads for Intermediate Computation with Language Models.” <https://arxiv.org/abs/2112.00114>.
- [4] Raj, Himanshu, Vivek Gupta, Davide Rosati, and Subrata Majumdar. 2023. “Semantic Consistency for Assuring Reliability of Large Language Models.” <https://arxiv.org/abs/2308.09138>.

Chapter 15

- [1] Bhatia, Kush, Anant Narayan, Christopher De Sa, and Christopher Ré. 2023. “TART: A Plug-and-Play Transformer Module for Task-Agnostic Reasoning.” <https://arxiv.org/abs/2306.07536>.
- [2] Lester, Brian, Rami Al-Rfou, and Noah Constant. 2021. “The Power of Scale for Parameter-Efficient Prompt Tuning.” In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 3045–3059. Association for Computational Linguistics.

Appendix A

- [1] Negro, Alessandro. 2021. *Graph-Powered Machine Learning*. Manning Publications.
- [2] Euler, Leonhard. 1736. “Solutio Problematis ad Geometriam Situs Pertinentis.” *Commentarii Academiae Scientiarum Imperialis Petropolitanae* 8: 128–40.
- [3] Diestel, Reinhard. 2017. *Graph Theory*. 5th ed. Springer.
- [4] Krebs, Valdis. 2012. “Political Book Networks.” *TNT: The Network Thinkers*, October. <http://www.thenetworkthinkers.com/2012/10/2012-political-book-network.html>.
- [5] Easley, David, and Jon Kleinberg. 2010. *Networks, Crowds, and Markets: Reasoning about a Highly Connected World*. Cambridge University Press.
- [6] Verbrugge, Lois M. 1977. “The Structure of Adult Friendship Choices.” *Social Forces* 56 (2): 576–597.
- [7] Cormen, Thomas H., Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein. 2009. *Introduction to Algorithms*. 3rd ed. MIT Press.

Appendix B

- [1] Robinson, Ian, Jim Webber, and Emil Eifrem. 2015. *Graph Databases*. 2nd ed. O’Reilly.
- [2] Misquitta, Luanne, and Christophe Willemsen. 2025. *Neo4j: The Definitive Guide*. O’Reilly.
- [3] Woodie, Alex. 2016. “Neo4j Pushes Graph DB Limits Past a Quadrillion Nodes.” *Datanami*, April 26. <https://www.datanami.com/2016/04/26/neo4j-pushes-graph-db-limits-past-quadrillion-nodes/>.

Appendix C

- [1] Lee, Rosalind C., Rhonda L. Feinbaum, and Victor Ambros. 1993. "The *C. elegans* Heterochronic Gene *lin-4* Encodes Small RNAs with Antisense Complementarity to *lin-14*." *Cell* 75 (5): 843–854.
- [2] Jonas, Stefanie, and Elisa Izaurrealde. 2015. "Towards a Molecular Understanding of microRNA-Mediated Gene Silencing." *Nature Reviews Genetics* 16 (7): 421–433.
- [3] Chen, Xing, et al. 2015. "RBMMMDA: Predicting Multiple Types of Disease-microRNA Associations." *Scientific Reports* 5 (1): 13877.
- [4] Bartel, David P. 2009. "MicroRNAs: Target Recognition and Regulatory Functions." *Cell* 136 (2): 215–233.
- [5] Miska, Eric A. 2005. "How microRNAs Control Cell Division, Differentiation and Death." *Current Opinion in Genetics & Development* 15 (5): 563–568.
- [6] Cheng, Angie M., et al. 2005. "Antisense Inhibition of Human miRNAs and Indications for an Involvement of miRNA in Cell Growth and Apoptosis." *Nucleic Acids Research* 33 (4): 1290–1297.
- [7] Cui, Qinghua, et al. 2006. "Principles of microRNA Regulation of a Human Cellular Signaling Network." *Molecular Systems Biology* 2 (1): 46.
- [8] Chen, Xing, Ming-Xi Liu, and Gui-Ying Yan. 2012. "RWRMDA: Predicting Novel Human microRNA–Disease Associations." *Molecular BioSystems* 8 (10): 2792–2798.
- [9] Chen, Xing, and Gui-Ying Yan. 2014. "Semi-Supervised Learning for Potential Human microRNA–Disease Associations Inference." *Scientific Reports* 4 (1): 5501.
- [10] Li, Jingjing, et al. 2012. "Evidence for Positive Selection on a Number of microRNA Regulatory Interactions during Recent Human Evolution." *PLoS Genetics* 8 (3): e1002578.
- [11] Chen, Kevin, and Nikolaus Rajewsky. 2006. "Natural Selection on Human microRNA Binding Sites Inferred from SNP Data." *Nature Genetics* 38 (12): 1452–1456.
- [12] Saunders, Matthew A., Han Liang, and Wen-Hsiung Li. 2007. "Human Polymorphism at microRNAs and microRNA Target Sites." *Proceedings of the National Academy of Sciences* 104 (9): 3300–3305.
- [13] Sethupathy, Praveen, and Francis S. Collins. 2008. "MicroRNA Target Site Polymorphisms and Human Disease." *Trends in Genetics* 24 (10): 489–497.
- [14] Deng, Su-Ping, Lin Zhu, and De-Shuang Huang. 2015. "Predicting Hub Genes Associated with Cervical Cancer through Gene Co-Expression Networks." *IEEE/ACM Transactions on Computational Biology and Bioinformatics* 13 (1): 27–35.
- [15] Deng, Su-Ping, Lin Zhu, and De-Shuang Huang. 2015. "Mining the Bladder Cancer-Associated Genes by an Integrated Strategy for the Construction and Analysis of Differential Co-Expression Networks." *BMC Genomics* 16 (3): 355.
- [16] Eulalio, Ana, Eric Huntzinger, and Elisa Izaurrealde. 2008. "Getting to the Root of miRNA-Mediated Gene Silencing." *Cell* 132 (1): 9–14.
- [17] Ambros, Victor. 2004. "The Functions of Animal microRNAs." *Nature* 431 (7006): 350–355.
- [18] Chen, Xing, et al. 2019. "MicroRNAs and Complex Diseases: From Experimental Results to Computational Models." *Briefings in Bioinformatics* 20 (2): 515–539.
- [19] Lukasik, Anna, Maciej Wójcikowski, and Piotr Zielenkiewicz. 2016. "Tools4miRs: One Place to Gather All the Tools for miRNA Analysis." *Bioinformatics* 32 (17): 2722–2724.
- [20] Lu, Ming, et al. 2008. "An Analysis of Human microRNA and Disease Associations." *PLoS ONE* 3 (10): e3420.
- [21] Li, Yang, et al. 2014. "HMDD v2.0: A Database for Experimentally Supported Human microRNA and Disease Associations." *Nucleic Acids Research* 42 (D1): D1070–D1074.
- [22] Huang, Zhou, et al. 2019. "HMDD v3.0: A Database for Experimentally Supported Human microRNA–Disease Associations." *Nucleic Acids Research* 47 (D1): D1013–D1017.

- [23] Xu, Feng, et al. 2022. “dbDEMC 3.0: Functional Exploration of Differentially Expressed miRNAs in Cancers of Human and Model Organisms.” *Genomics, Proteomics & Bioinformatics* 20 (3): 446–454.
- [24] Jiang, Qinghua, Yadong Wang, Yangyang Hao, Liran Juan, Mingxiang Teng, Xinjun Zhang, et al. 2009. “miR2Disease: A Manually Curated Database for microRNA Deregulation in Human Disease.” *Nucleic Acids Research* 37 (suppl_1): D98–D104.
- [25] Bodenreider, Olivier. 2004. “The Unified Medical Language System (UMLS): Integrating Biomedical Terminology.” *Nucleic Acids Research* 32 (suppl_1): D267–D270.
- [26] Kozomara, Ana, Maria Birgaoanu, and Sam Griffiths-Jones. 2019. “miRBase: From microRNA Sequences to Function.” *Nucleic Acids Research* 47 (D1): D155–D162.
- [27] Kozomara, Ana, and Sam Griffiths-Jones. 2014. “miRBase: Annotating High Confidence microRNAs Using Deep Sequencing Data.” *Nucleic Acids Research* 42 (D1): D68–D73.
- [28] Kozomara, Ana, and Sam Griffiths-Jones. 2010. “miRBase: Integrating microRNA Annotation and Deep-Sequencing Data.” *Nucleic Acids Research* 39 (suppl_1): D152–D157.
- [29] Griffiths-Jones, Sam, et al. 2006. “miRBase: microRNA Sequences, Targets and Gene Nomenclature.” *Nucleic Acids Research* 34 (suppl_1): D140–D144.
- [30] Chen, Yuhao, and Xiaowei Wang. 2020. “miRDB: An Online Database for Prediction of Functional microRNA Targets.” *Nucleic Acids Research* 48 (D1): D127–D131.
- [31] Liu, Weijun, and Xiaowei Wang. 2019. “Prediction of Functional microRNA Targets by Integrative Modeling of microRNA Binding and Target Expression Data.” *Genome Biology* 20 (1): 18.
- [32] Kern, Fabian, Ernesto Aparicio-Puerta, Yongping Li, Tobias Fehlmann, Tim Kehl, Viktoria Wagner, et al. 2021. “miRTargetLink 2.0—Interactive miRNA Target Gene and Target Pathway Networks.” *Nucleic Acids Research* 49 (W1): W409–W416.
- [33] Li, Jianwei, Shan Zhang, Yanping Wan, Yingshu Zhao, Jiangcheng Shi, Yuan Zhou, and Qinghua Cui. 2019. “MISIM v2.0: A Web Server for Inferring microRNA Functional Similarity Based on microRNA–Disease Associations.” *Nucleic Acids Research* 47 (W1): W536–W541.
- [34] Spalinger, Marianne R., Declan F McCole, Gerhard Rogler, and Michael Scharl. 2015. “Role of Protein Tyrosine Phosphatases in Regulating the Immune System: Implications for Chronic Intestinal Inflammation.” *Inflammatory Bowel Diseases* 21 (3): 645–655.
- [35] Yao, Ying, Łukasz Wyrożemski, Knut E A Lundin, Geir Kjetil Sandve, and Shuo-Wang Qiao. 2021. “Differential Expression Profile of Gluten-Specific T Cells Identified by Single-Cell RNA-Seq.” *PLoS ONE* 16 (10): e0258029.

